

Session 1 to 5

Session 1

1. Name 5 IT job roles and write one line about what each role does.

1. **Software Developer** – Writes and maintains software applications.
 2. **Web Developer** – Builds and maintains websites and web applications.
 3. **Software Tester (QA Engineer)** – Tests software to find bugs and ensure quality.
 4. **Database Administrator (DBA)** – Manages and secures databases.
 5. **System Administrator** – Maintains computer systems, servers, and networks.
-

2. List 3 product companies and 3 service companies with one example of work each does.

Product Companies

1. **Microsoft** – Develops the Windows operating system and Microsoft Office.
2. **Google** – Creates products like Android and Google Chrome.
3. **Adobe** – Develops software such as Photoshop and Acrobat.

Service Companies

1. **TCS (Tata Consultancy Services)** – Provides software development and IT consulting services.
 2. **Infosys** – Offers application development and digital transformation services.
 3. **Wipro** – Provides cloud computing, cybersecurity, and IT support services.
-

3. Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.

Term	Full Form	Use
GitHub	Git Hub (Git-based hosting platform)	Stores and manages code using Git version control.
Slack	<i>Slack is not an acronym</i>	Used for team communication and collaboration.
Jira	<i>Jira is not an acronym</i>	Used for project management and bug tracking.
IDE	Integrated Development Environment	Used to write, edit, debug, and run code.

API	Application Programming Interface	Allows different software applications to communicate with each other.
------------	-----------------------------------	--

4. Name any 5 programming languages and mention one popular use case for each.

1. **Python** – Data science and machine learning.
2. **Java** – Android application development.
3. **C++** – Game development and system programming.
4. **JavaScript** – Creating interactive websites.
5. **C#** – Developing Windows applications and Unity games.

5. What is the difference between a developer and a tester? (2 lines)

- A **Developer** writes and maintains software code to build applications.
- A **Tester** checks the software for bugs and ensures it works correctly before release.

Session 2

1. What are the three main layers of software architecture? Write one line about each.

1. **Presentation Layer (UI Layer)** – Displays information to users and accepts their input.
2. **Business Logic Layer (Application Layer)** – Processes data and implements the application's rules and logic.
3. **Data Layer (Database Layer)** – Stores, retrieves, and manages data in the database.

2. Define client and server in your own words with one real-life example.

- **Client:** A client is a device or application that requests services or data from a server.
- **Server:** A server is a computer or system that provides data or services to clients.

Example: When you open YouTube on your phone, your phone is the **client**, and YouTube's computers that send videos are the **servers**.

3. Write 2 differences each between desktop apps, web apps, and mobile apps.

Desktop Apps	Web Apps	Mobile Apps
--------------	----------	-------------

Installed on a computer.	Run in a web browser.	Installed on a smartphone or tablet.
Can work without the internet (in many cases).	Usually require an internet connection.	Can use mobile features like GPS and camera.

4. Name any 3 databases and mention whether each is SQL or NoSQL.

1. **MySQL – SQL**
2. **PostgreSQL – SQL**
3. **MongoDB – NoSQL**

Session 3

1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Term	Full Form	One-line Use
HTTP	HyperText Transfer Protocol	Used to transfer web pages between a browser and a web server.
HTTPS	HyperText Transfer Protocol Secure	Secure version of HTTP that encrypts data during communication.
FTP	File Transfer Protocol	Used to upload and download files between computers over a network.
SMTP	Simple Mail Transfer Protocol	Used to send emails from one mail server to another.

2. What is the difference between IPv4 and IPv6? (2 lines)

- **IPv4** uses **32-bit** addresses (e.g., **192.168.1.1**) and supports fewer unique IP addresses.
- **IPv6** uses **128-bit** addresses (e.g., **2001:db8::1**) and supports a much larger number of unique IP addresses.

3. Run `nslookup google.com` on your system and write down the IP address you get.

I can't run commands on **your computer**, so I cannot see the exact result.

You can run:

```
1 nslookup google.com
2
```

Example output (may vary by location):

```
1 google.com
2 Address: 142.250.183.14
3
```

Note: Google uses multiple servers, so the IP address you receive may be different.

4. What do these HTTP status codes mean?

- **200 OK** – The request was successful.
 - **301 Moved Permanently** – The requested page has been permanently moved to a new URL.
 - **404 Not Found** – The requested page or resource could not be found.
 - **500 Internal Server Error** – The server encountered an unexpected error.
 - **403 Forbidden** – The server understood the request but denied access.
-

5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

- **Root (.)** – The top level of the DNS hierarchy (usually hidden).
- **TLD (Top-Level Domain)** – **.com**
- **Domain** – **google**
- **Subdomain** – **www**

Example:

`www.google.com`

- **www** → Subdomain
- **google** → Domain Name
- **.com** → Top-Level Domain (TLD)
- **.(root)** → Root Domain (implicit and usually not shown)

Session 4

1. List all 6 phases of SDLC in order with one line about each.

1. **Requirement Analysis** – Gather and understand the customer's requirements.
 2. **System Design** – Create the architecture and design of the software.
 3. **Development (Coding)** – Write the source code based on the design.
 4. **Testing** – Test the software to find and fix bugs.
 5. **Deployment** – Release the software to users or the production environment.
 6. **Maintenance** – Update, improve, and fix issues after deployment.
-

2. Write 3 differences between Agile and Waterfall models.

Agile	Waterfall
Development is done in small iterations (Sprints).	Development is completed in one sequential process.
Changes can be made easily during the project.	Changes are difficult after a phase is completed.
Customer feedback is taken regularly.	Customer feedback is usually taken after the project is finished.

3. Name the 3 main Scrum roles and write one line about the responsibility of each.

1. **Product Owner** – Defines product requirements and prioritizes the work.
 2. **Scrum Master** – Helps the team follow Scrum practices and removes obstacles.
 3. **Development Team** – Designs, develops, tests, and delivers the product.
-

4. What is a Sprint in Scrum? Mention its typical duration.

A **Sprint** is a fixed time period in Scrum during which the team completes a set of planned tasks and delivers a working product increment.

Typical Duration: 1 to 4 weeks (most commonly **2 weeks**).

5. What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

- **Backlog** – A list of all pending tasks or user stories that are not yet started.
- **To Do** – Tasks that are ready to be started.

- **In Progress** – Tasks that are currently being worked on.
- **Done** – Tasks that have been completed successfully.

Session 5

1. Write the Git command for each.

Task	Git Command
Initialize a repository	<code>git init</code>
Stage all files	<code>git add .</code>
Commit with a message	<code>git commit -m "Initial commit"</code>
Push to remote	<code>git push origin main</code>
Pull latest changes	<code>git pull origin main</code>

2. What is the difference between `git fetch` and `git pull` ? (2 lines)

- `git fetch` downloads the latest changes from the remote repository but does **not** merge them into your current branch.
- `git pull` downloads the latest changes **and automatically merges** them into your current branch.

3. Create a new GitHub repository, push any small text file from your local system, and share the repo link.

I can't create a GitHub repository or push files on your behalf.

You can do it using these commands:

```
1 git init
2 echo "Hello GitHub" > demo.txt
3 git add .
4 git commit -m "First commit"
5 git branch -M main
6 git remote add origin https://github.com/YOUR_USERNAME/YOUR_REPO.git
7 git push -u origin main
8
```

After pushing, share the repository link in this format:

```
1 https://github.com/YOUR_USERNAME/YOUR_REPO
2
```

4. What is the difference between a Fork and a Pull Request? (2 lines)

- A **Fork** is a personal copy of someone else's GitHub repository in your own GitHub account.
 - A **Pull Request (PR)** is a request to merge your changes into the original repository.
-

5. Create a new branch named `feature-test`, make one commit on it, and merge it back to the `main` branch. Write the commands used.

```
1 git checkout -b feature-test
2
3 echo "New feature" > feature.txt
4
5 git add .
6 git commit -m "Add feature file"
7
8 git checkout main
9
10 git merge feature-test
11
```

These are the standard Git commands you would use to complete the task.